

Al-Balqa' Applied University
Faculty of Engineering Technology



Electrical Engineering Department

Laboratory Manual

For

Electric Circuit

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Series-parallel Dc circuits

Experiment
DC

1

Object:

To investigate the characteristics of series-parallel dc circuits.

Equipment Required:

Resistors:

91 Ω,220 Ω, 330 Ω, 470Ω

Instruments:

1-- DMM or VOM

1-- DC power supply

#EQUIPMENT ISSUED

TABLE 1.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Power supply		

TABLE 1.2

Resistors	
Nominal value	Measured value
91Ω	
220Ω	
330Ω	
470Ω	

RESUME OF THEORY

The analysis of series –parallel dc networks requires a firm understanding of the basics of both series and parallel networks. In the series-parallel configuration, you will have to isolate series and parallel configurations and make the necessary combinations for reduction as you work toward the desired unknown quantity.

As a rule, it is best to make a mental sketch of the path you plan to take toward the complete solution before introducing the numerical values this may result in a savings in both time and energy, always work with the isolated series or parallel combinations in a branch before trying the branches together, for complex networks carefully redrawn set of reduced networks may be required to ensure that the unknown is conserved and that every element has been properly included.

Procedure:

PART 1

(a) Construct the series-parallel network of fig 1.1 insert the measured resistor value next to each resistor.

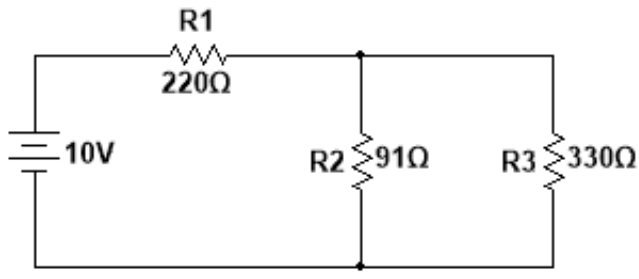


FIG 1.1

(b) Calculate the total resistance R_T using the measured resistance values.

$R_{T(CALCULATED)}$ =———————— .

(c) Use the ohmmeter Section of the Multimeter to measure R_T and compare to the result of (b)

$R_{T(MEASURED)}$ =————————.

(d) If 10V were applied to points 1-2, calculate the currents I_1, I_2 and I_3 using the measured resistor values

$I_{1(CALCULATED)}$ =————————, $I_{2(CALCULATED)}$ =————————, $I_{3(CALCULATED)}$ =————————.

(e) Apply 10V to points 1-2 and measure the currents I_1, I_2 and I_3 using the multimeter section of your multimeter.

$I_{1(MEASURED)}$ =————————, $I_{2(MEASURED)}$ =————————, $I_{3(MEASURED)}$ =————————,

(f) Using the results of 4, calculate the voltages V_1, V_2 and V_3.

V_1=_____, V_2=_____, V_3=_____

PART 2

(a) Construct the series-parallel network OF Fig 1.2, insert the measured value of each resistor next to the nominal color code.

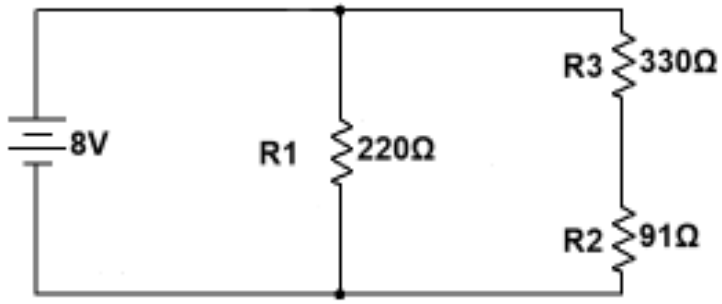


FIG 1.2

(b) Calculate the total resistance R_T using the measured value of each resistor.

$R_{T(CALCULATED)}$ =————————.

(c) Use the ohmmeter section of your multimeter to measure the total resistance R_T compare to the result of (b).

$R_{T(MEASURED)}$ =————————.

(d) If 10V were applied to points 1-2 calculate the currents I_T, I_1, I_2 and I_3.

I_T =————————, I_1=————————, I_2=————————, I_3=————————.

(e) Apply 10V to points 1-2 and measure the currents I_T, I_1, I_2 and I_3. compare to the results of (d).

$I_{T(MEASURED)}$ =————————, $I_{1(MEASURED)}$ =————————,

$I_{2(MEASURED)}$ =————————, $I_{3(MEASURED)}$ =————————.

(f) Using the results of (d), calculate the voltages V_1, V_2 and V_3.

V_1 =————————, V_2 =————————, V_3 =————————.

(g) Measure the voltages V_1, V_2 and V_3, compare to the results of (f).

V_1 =————————, V_2 =————————, V_3 =————————.

Part 3

(a) Construct the series-parallel network of Fig 1.3 and insert the measured values of each resistor.

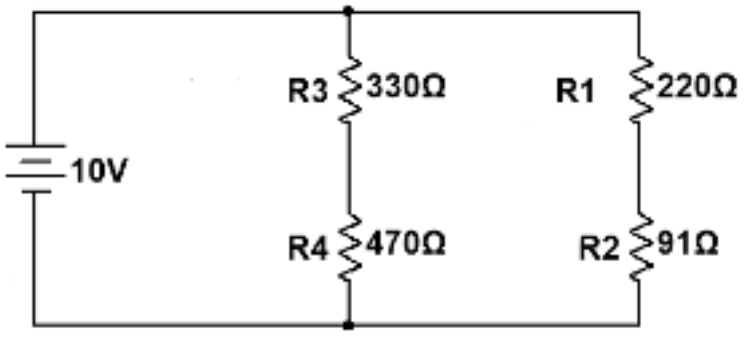


FIG 1.3

(b) Using the measured values calculate the voltages V_{ad} and V_{bc}.

V_{ad} =————————, V_{bc} =————————.

(c) Measure the voltages V_{ad} and V_{bc} and compare to the results of (b).

V_{ad} =————————, V_{bc} =————————.

(d) Using the results of (b), calculate the voltage V_{ab}.

$V_{ab(CALCULATED)}$ =————————.

(e) Measure the voltages V_{ab} and compare to the results of (d).

$V_{ab(MEASURED)}$ =————————.

PART 4

(a) Construct the network of Fig 1.4 insert the measured values of each resistor.

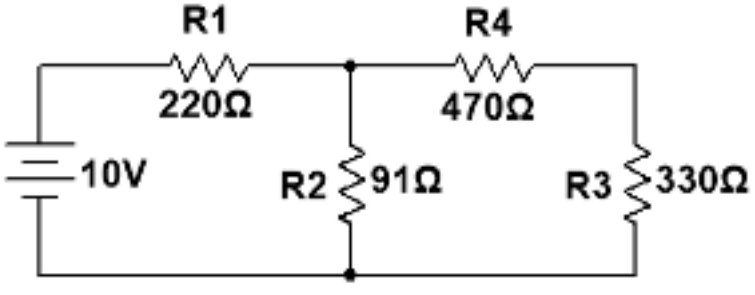


FIG 1.4

(b) Calculate the voltage of V4 using the resistor values.

$V_{4(CALCULATED)}$ =————————————.

(c) measure the voltage of V4 and compare to the result of (b).

$V_{4(MEASURED)}$ =————————————.

METHODS OF ANALYSIS

Experiment
DC

2

OBJECT:

To become familiar with the branch mesh and nodal-analysis techniques.

EQUIPMENT REQUIRED

RESISTORS:

1.2KΩ, 2.2KΩ, 3.3KΩ

INSTRUMENTS

1-DMM OR VOM

2-DC POWER SUPPLIES

#EQUIPMENT ISSUED

TABLE 2.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Power supply		
Power supply		

TABLE 2.2

Resistors	
Nominal value	Measured value
1.2KΩ	
2.2KΩ	
2.2KΩ	
3.3KΩ	
3.3KΩ	

RESUME OF THEORY

The branch mesh and nodal analysis techniques are used to solve complex networks with a single source or networks with more than one source that are not in series or parallel.

The brand and mesh analysis techniques will determine the currents of the network, while nodal analysis approach will provide the potential levels of the nodes of the network with respect to some reference.

The application of each techniques follows a sequence of steps each of which will resulting a set of equations with the desired unknowns, an application of determinants or other mathematical procedures will then provide the results required.

For all the difference calculations the equation is:

%Difference= [Measured-Calculated] / Calculated * 100%

Procedure:

Part 1:

(a) Construct the network of fig 2.1. insert the measured values of the resistors:

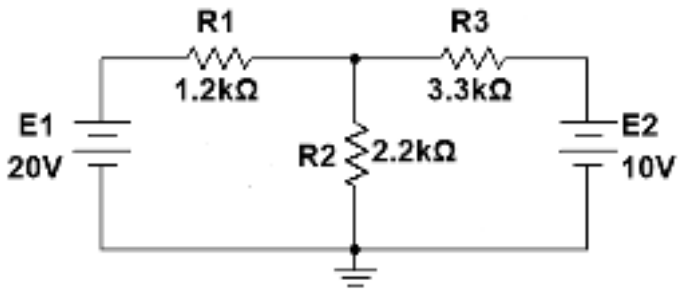


FIG 2.1

(b) Using branch-current analysis, calculate the current through each branch of the network of Fig 2.1 and insert in Table 2.3, use the measured resistor values and assume the current direction shown in figure:

TABLE 2.3

Current	Calculated	Measured	Difference%
I_1			
I_2			
I_3			

(c) Measure the voltages V_1, V_2 and V_3 and enter below with minus signs if the polarity is opposite to that of Fig 2.1:

V_1=_____, V_2=_____, V_3=_____.

(d) Calculate the current I_1, I_2 and I_3 and insert in Table 2.3, as the measured values, be sure to include minus signs the polarity is opposite to that of Fig 2.1, calculate the difference for each current:

(e) Using nodal analysis determine the nodal voltage for the network of Fig 2.1 use the measured resistor values and be sure to convert each source to current source before applying the method.

V_n =_____.

(f) Since V_n =v3 insert the measured value of v3 below :

V_n =V_3 =_____.

(g) Calculate the difference between 2 values:

Difference% =_____.

Part 2:

(a) Construct the network, insert the measured value of each resistor in the diagram:

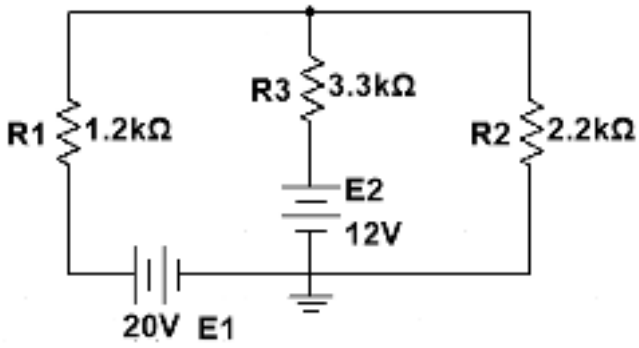


FIG 2.2

(b) Using mesh analysis, determine the current through each resistor, use the measured resistor values and the indicated direction for the mesh currents, insert the calculated resistor currents in table:

Mesh currents

I_1 =————————, I_2 =————————.

TABLE 2.4

Current	Calculated	Measured	Difference%
I_1			
I_2			
I_3			

(c) Measure the voltages V_1, V_2 and V_3 and enter below with minus signs if the polarity is opposite to the fig 2.2:

V1 =————————, V2 =————————, V_3 =————————.

(d) Calculate the currents I_{R1}, I_{R2} and I_{R3} and insert in table 2.4 as measured values be sure to include minus signs in the table if the current direction is opposite, calculate the percent difference for each current.

(e) Using nodal analysis determine the nodal voltage for the network of fig 2.2 use the measured resistor values

V_N =————————————.

(f) Since V_N=V_2 insert the measured value of V_2 below:

V_N=V_2 =————————————.

(g) Calculate the difference between 2 values:

%Difference =————————————.

PART 3

(a) Construct the network of FIG 2.3 insert the measured resistor values

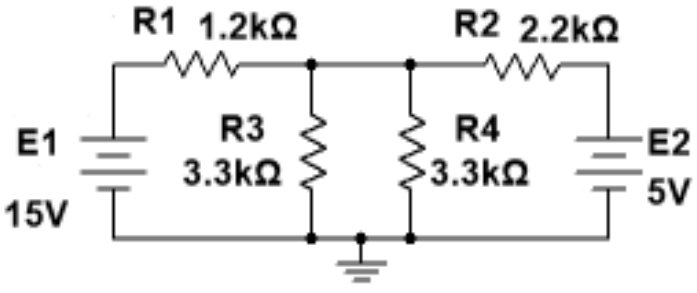


FIG 2.3

Caution: Be sure DC power supplies are hooked up as shown (common ground) before turning the power on

(b) Using nodal analysis calculate the voltage using measured resistor values

$V_{A(CALCULATED)}$ =————————.

(c) Measure the voltage V_A and record below

$V_{A(MEASURED)}$ =————————.

(d) Calculate the percent difference between 2 values

%Difference =————————.

Part 4

(a) Construct the network of Fig 2.4 insert the measured resistor values.

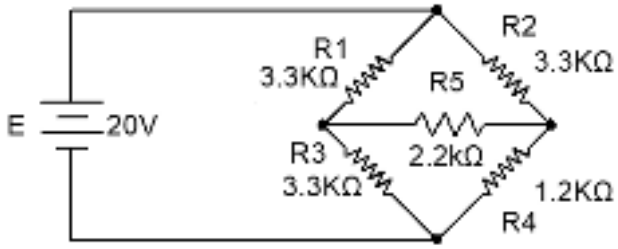


FIG 2.4

(b) Using any one of the three techniques examined in this experiment, calculate the voltage V_5 and the current I_5, use the measured resistor values.

$V_{5(calculated)}$ =————————, $I_{5(calculated)}$ =————————.

(c) Measure the voltage V_5 and insert below with a minus sign if the polarity is different from that appearing in Fig 2.4.

$V_{5(measured)}$ =————————.

(d) Calculate the percent difference between the two values.

%Difference =————————.

Superposition Principle (DC)

Experiment
DC

3

OBJECT

To verify experimentally the superposition principle as applied to dc circuits.

EQUIPMENT REQUIRED

Resistors

1-1.2KΩ, 2.2KΩ, 3.3KΩ

Instruments

1- DMM or VOM

2-DC power supplies

#EQUIPMENT ISSUED

TABLE 3.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Power supply		
Power supply		

TABLE 3.2

Resistors	
Nominal value	Measured value
1.2KΩ	
2.2KΩ	
3.3KΩ	

RESUME OF THEORY

The superposition principle states that the current through, or voltage across, any resistive branch of a multisource network is the algebraic sum of the contributions due 10 each source action independently, when the effect of one source are considered, the others are replaced by their internal resistance, super position is effective only for linear circuit relationships.

The principle permits one to analyze circuits without resorting to simultaneous equation, nonlinear effects such as power which varies as the square of the current or voltage cannot be analyzed using this principle.

PROCEDURE

PART 1

(a) Construct the network of Fig 3.1 and insert the measured resistor values.

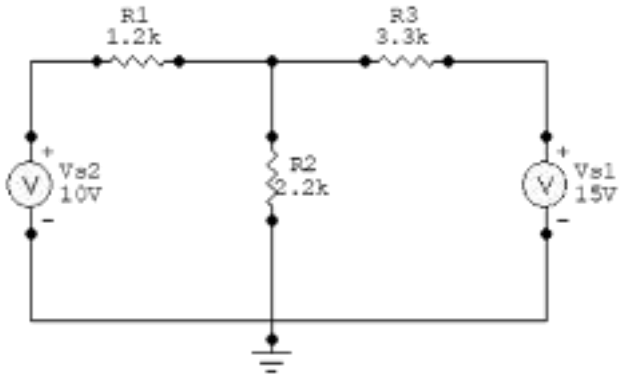


FIG 3.1

Caution: Be sure DC power supplies are hooked up as shown (common ground) before turning the power on

(b) Using superposition and the measured resistor values, calculate the currents indicated in table 3.3 for the network of Fig 3.1 next to each magnitude include a small arrow to indicate the current direction for the complete network.

TABLE 3.3 calculated values for the network of fig 3.1

Due to E_1	Due to E_2	Algebraic Sum
$I_1' =$	$I_1'' =$	$I_1 =$
$I_2' =$	$I_2'' =$	$I_2 =$
$I_3' =$	$I_3'' =$	$I_3 =$

(c) Energize the network for Fig 3.1 and measure the voltages indicated in Table 3.4 calculated the currents in Table 3.4 using ohm's law indicate the polarity of the voltages and the direction of the currents on Fig 3.1

TABLE 3.4 Measured values for the network of Fig 3.1

V_1 =
V_2 =
V_3 =
I_1 =
I_2 =
I_3 =

(d) Compare the results of (b) and (c)

(e) Construct the network of Fig 3.2 note: The source (E2) has been removed.

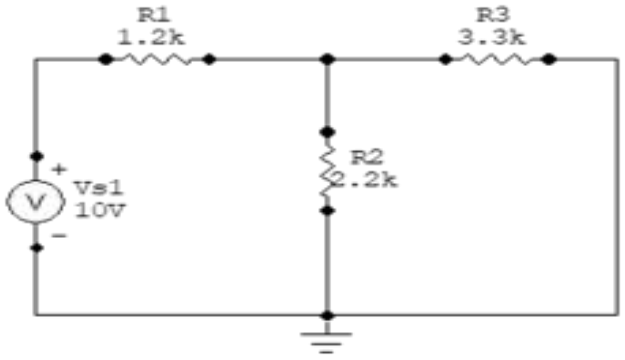


FIG 3.2

(f) energize the network of fig 3.2 and measure the voltages indicated in table 3.5 calculate the currents using ohm's law use measured resistor values.

TABLE 3.5 Measured values for the network of Fig 3.2

V_1 =
V_2 =
V_3 =
I_1 =
I_2 =
I_3 =

(g) Calculated the percent difference between calculated and measured (as determined from the voltage reading) values for Fig 3.2 using the calculated results of Table 3.3 and the equation:

$$\%\ Difference = \frac{\left|I_{(Measured)} - I_{(Calculated)}\right|}{I_{(calculated)}} * 100\% \qquad (Eq\ 3.1)$$

%Difference (I_1) =————, %Difference (I_2) =————, %Difference (I_3) =————.

(h) Construct the network of Fig 3.3 note the source (E1) has been removed.

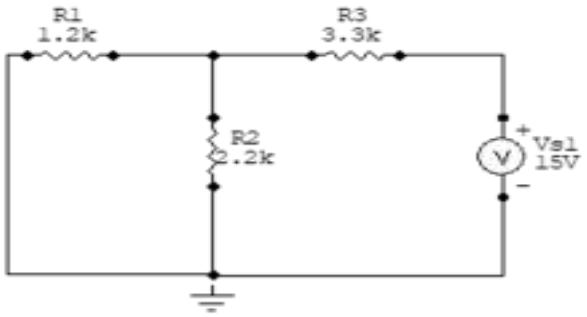


FIG 3.3

(i) Energize the network of Fig 3.3 and measure the voltage indicated in table 3.6 calculated the currents using Ohm's law

TABLE 3.6 Measured values for the network of Fig 3.3

V_1 =
V_2 =
V_3 =
I_1 =
I_2 =
I_3 =

(j) Calculate the percent difference calculated and measured value for Fig 3.3 use the calculated results of 3.3 and E_q (3.1)

%Difference (I3) =————————, %Difference (I2) =————————, %Difference (I1) =————————.

(k) Insert the algebraic sum of the calculate and measured current in table 3.7 insert a small arrow to indicate the resultant direction.

TABLE 3.7 Algebraic results of the networks of Fig 3.1

Calculated (from Table 3.3)	Measured (from (f) and (i))
I_1 =	I_1 =
I_2 =	I_2 =
I_3 =	I_3 =

(l) calculate the percent difference between the measured values of table 3.7 and 3.4 using the following equation

$$\%\, Difference = \frac{|I_{Table\ 3.7} - I_{Table\ 3.4}|}{I_{Table\ 3.4}} * 100\% \qquad (Eq\ 3.2)$$

%Difference (I3) =————, %Difference (I2) =————, %Difference (I1) =————.

(m) Using the result of (f), determine the power delivered to each resistor and insert in Table 3.8 repeat for the results of (i), finally include the results obtained using the measurements of (c) and complete the Table.

TABLE 3.8

(f)	(i)	Sum of (f) and (i)	(c)
$P_1^{\backslash}$ =	$P_1^{\backslash\backslash}$ =	P_1 =	P_1 =
$P_2^{\backslash}$ =	$P_2^{\backslash\backslash}$ =	P_2 =	P_2 =
$P_3^{\backslash}$ =	$P_3^{\backslash\backslash}$ =	P_3 =	P_3 =

(n) referring to the results of (1), Comment on the applicability of the superposition theorem to power effects.

Part 2

(a) construct the network of fig 3.4 insert the measured resistor values.

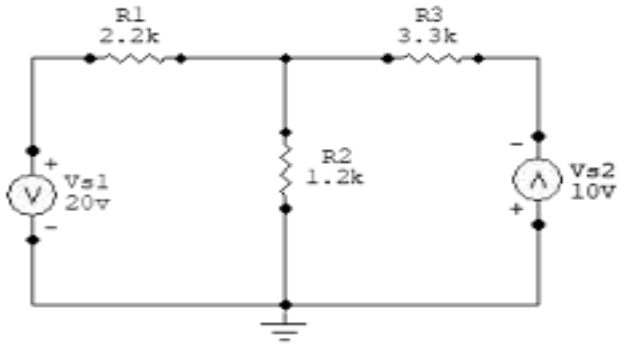


FIG 3.4

(b) using the measured resistor, calculate the voltage V_1, V_2 and I using superposition show your work in the space provided

$V_{1=}$_____, $V_{2=}$_____, $V_{3=}$_____

(c) Energize the network of Fig 3.4 and measure the voltage V_1, V_2, V_3 is the superposition principle verified?

$V_{1=}$_____, $V_{2=}$_____, $V_{3=}$_____

(d) using measured resistor value, calculate the voltage v2 due only to source (E1). modify the network and measured the voltage V_2 and only to source (E1), be sure to indicate the polarity of V_2.

$V_{1(Calculated)=}$_____, $V_{2(Measured)=}$_____.

(e) repeat (d)for source (E2).

$V_{1(Calculated)=}$_____, $V_{2(Measured)=}$_____.

(f) using the measured results of d and r, determine the voltage V_2 and compare to the result of (c). calculate the present difference using the following

$$\%\ Difference = \frac{\left|V_{2(from\ (f))} - V_{2(from\ (c))}\right|}{V_{2(from\ (c))}} * 100\% \qquad (Eq\ 3.3)$$

$V_{2(from\ (d)\ and\ (e))}$ =————————, $V_{2(from\ (c))}$ =————————, %Difference =————————.

Theremin's Theorem and Maximum Power Transfer

Experiment DC

4

#OBJECT

To verify Theremin's theorem and the Maximum power transfer principle.

EQUIPMENT REQUIRED

Resistors
91Ω, 220Ω, 330Ω, 470Ω.
(0-1) KΩ Potentiometer

Instruments
DMM or VOM
DC Power supply

#EQUIPMENT ISSUED

TABLE 1.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Power supply		

TABLE 1.2

Resistors	
Nominal value	Measured value
91Ω	
220Ω	
330Ω	
470Ω	

RESUME OF THEORY

Through the use of Thevenin's theorem, a complex two terminal, linear, multisource DC network can be replaced by one having a single source and resistors.

The Thevenin's equivalent circuit consists of a single dc source referred to as Thevenin's voltage is the open circuit voltage across the terminals.

In question, the Thevenin's resistance is the resistance between these terminals with all of the voltage and current sources replaced by their internal resistance.

If maximum power is delivered by the dc source to the resistor, the resistance must have a same value of Thevenin's equivalent resistance.

$$P_{max} = \frac{{E_{TH}}^2}{4R_{TH}}$$

PROCEDURE

Part 1: Thevenin's theorem Calculations

(a) Insert the measured resistor values into Fig (4.1) and calculate the Thevenin's voltage.

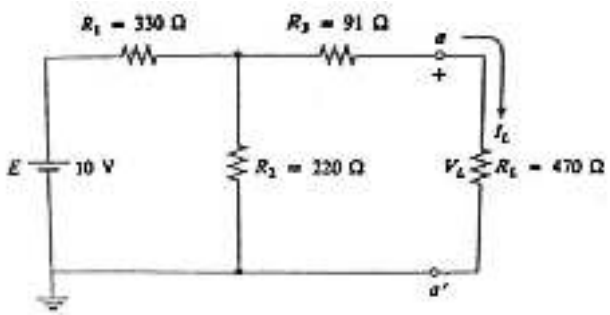


FIG 4.1

E_{TH} =————————————, R_{TH} =————————————.

(b) Insert the values obtained in (a) in the schematic of fig 4.2 and calculate the load current.

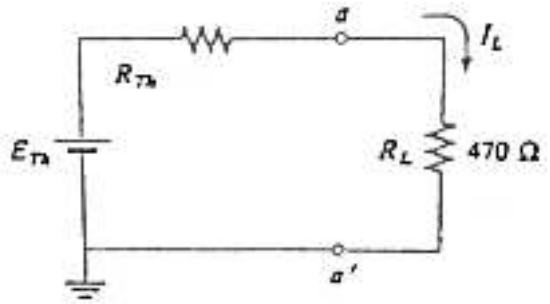


FIG 4.2

I_L =————————————.

(c) calculate the load current in the original network of fig 4.1 using series parallel methods

I_L =————————————.

Measurements:

(d) Construct the network of Fig 4.1 and measure the voltage for the load and the current for the load and compare to (b).

$V_{L\ (measured)}$ =————————————, $I_{L\ (Calculated\ from\ VL)}$ =————————————.

(e) Determine R_{Th} by constructing the network of Fig 4.3 and measuring the resistance between the DMM terminals.

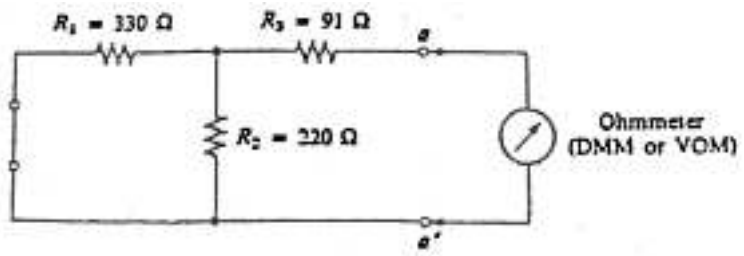


FIG 4.3

R_{Th} =————————————.

How does it compare to the value determine in (a)? Calculate the percent difference from

$$\%\ Difference = \frac{\left|R_{Th(from\ (e))} - R_{Th(from\ (a))}\right|}{R_{Th(from\ (a))}} * 100\% \qquad (Eq\ 4.2)$$

%Difference =————————.

(f) Determine E_{th} by constructing the network of Fig 4.4 and measuring the open circuit voltage between the load's terminals.

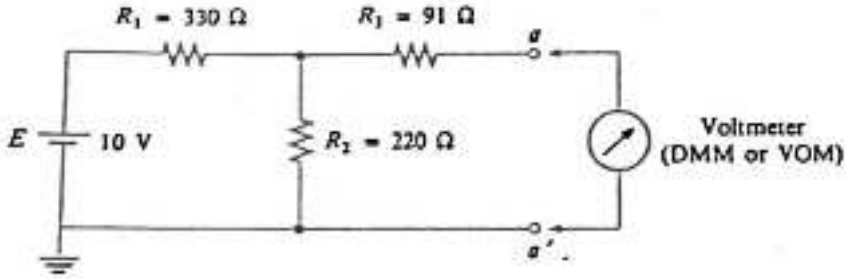


FIG 4.4

E_{Th} =————————————.

How does this value compare to the result of (a)? Determine the present difference using an equation similar to that appearing in (e).

%Difference =————————.

Thevenin's Network:

(g) Construct the network of Fig 4.5 and set the values obtained for E_{Th} and R_{Th} in (e) and (f), respectively, use the ohmmeter section of your meters to set the potentiometer properly, then measure the voltage V_L and calculate the current I_L.

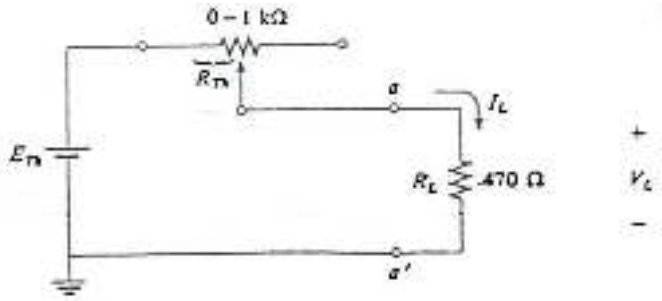


FIG 4.5

$V_{L\ (measured)}$ =————————————, $I_{L\ (Calculated\ from\ VL)}$ =————————————.

Compute the percent difference between levels of I_L from (b) and the above using:

$$\%\ Difference = \frac{\left|I_{L(from\ (g))} - I_{L(from\ (b))}\right|}{I_{L(from\ (b))}} * 100\% \qquad (Eq\ 4.3)$$

%Difference =————————.

Part 2 Maximum Power Transfer:

(a)Construct the network of fig 4.6 and set the potentiometer to 50 measure the voltage across R as you vary R through the following value 50, 100, 200, 300, 330, 400, 600, 800 and 1000Ω, be sure to set the resistance with the ohmmeter section of your meter before each reading (remember to disconnect the dc supply when setting the resistance level), complete Table 4.3 and plot power versus R.

TABLE 4.3

R	V_R	$P_R = V_R{}^2/R$
50Ω		
100Ω		
200Ω		
300Ω		
R_L (Measured)		
400Ω		
600Ω		
800Ω		
1KΩ		

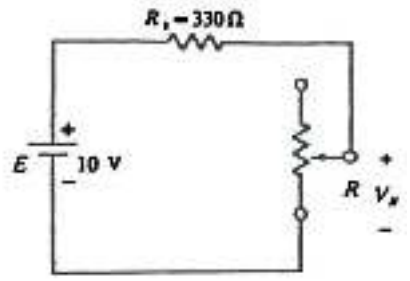
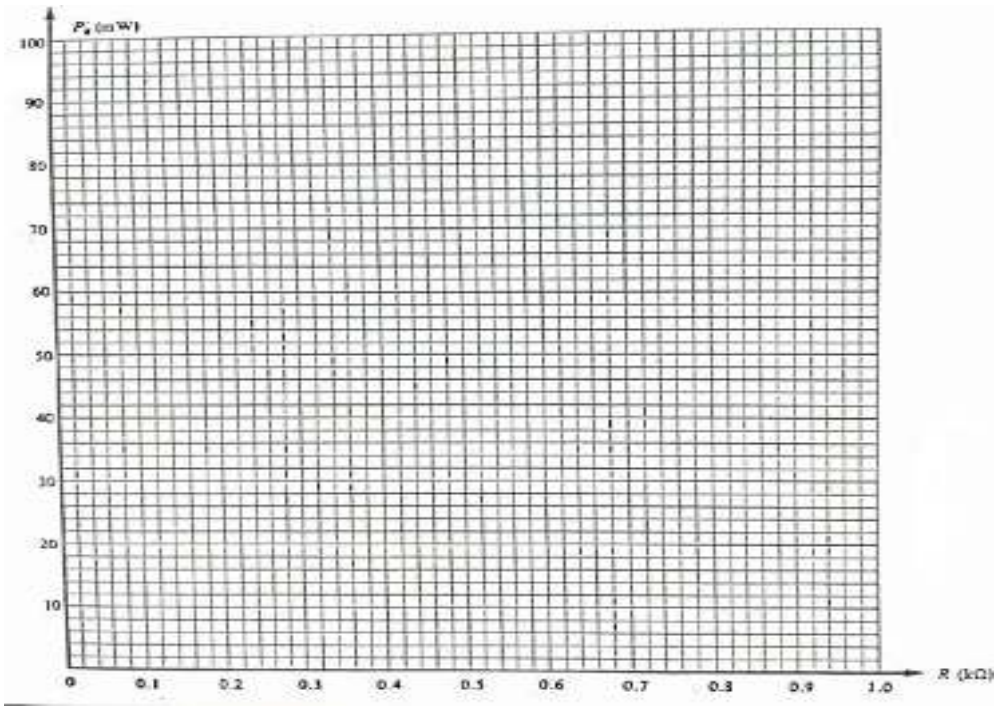


FIG 4.6



(b) Theoretically, for what value of R will the power delivered to R be maximum? check this value against that obtained from the graph.

R =_____.

In addition what should the voltage across R be when R is set for maximum power? do your experiment data substantiate this?

V_R =_____.

(c) Construct the network of Fig 4.7 and set the potentiometer to 10Ω complete Table 4 4 for each value of R.

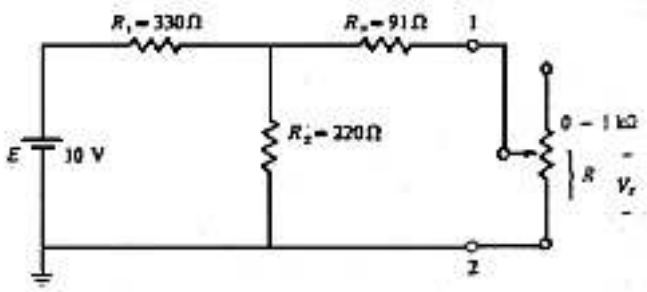


FIG 4.7

TABLE 4.4

R	V_R	P_R
10		
50Ω		
100Ω		
150Ω		
200Ω		
250Ω		
300Ω		
350Ω		
400Ω		

(d) What is the value of R would appear to result in a maximum power transfer to the load? How does it compare to the value of R_{Th} determined in Part 1 (a)?

R =————————————.

Part 3 Concluding Exercise

(a) Construct the network of gig 4.8, insert the measured resistor values.

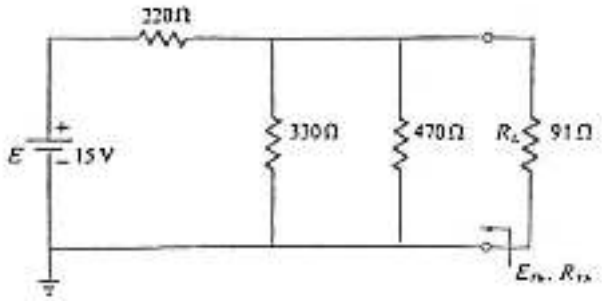


FIG 4.8

(b) Using experimental techniques, determine the maximum power to the resistor R_L, that is, determine E_{Th} and R_{Th} experimentally and then use Eq (4.1)

E_{Th} =————————————, R_{Th} =————————————, P_{max} =————————————.

(c) Calculate the value of E_{Th}, R_{Th} and P_{max} using measured resistor value and compare to the results of (b) , using Table 4.5

TABLE 4.5

Parameter	(b)	(c)	$\%Difference = (\lvert(c) - (b)\rvert/(b)) * 100\%$
E_{Th}			
R_{Th}			
P_{max}			

(d) Comment on whether R_L =91Ω is a poor or excellent choices for maximum power transfer to the load, with R_L =91Ω, what percent of maximum power is delivered to the load using the equation ($P_{91\Omega}/P_{max}$) *100%?

%of maximum power =————————————.

Wheatstone bridge and ∆-Y Conversions

Experiment DC

5

#OBJECT

To become familiar with the Wheatstone bridge and ∆-Y conversions.

EQUIPMENT REQUIRED

Resistors

1--91Ω

2--200Ω

3--330Ω,1KΩ

1--(0-1) KΩ potentiometer

1--Unmark fixed resistor in the range 47Ω to 220Ω

Instruments

1—DMM or VOM

1—DC Power supply

1—Commercial Wheatstone bridge (if available)

#EQUIPMENT ISSUED

TABLE 5.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Power supply		
Wheatstone bridge		

Since more than one resistor of a particular value will appear in some configurations of this experiment, the measured value of each should be determined just prior to inserting them into the network.

#**RESUME OF THEORY**

The Wheatstone bridge is an instrument used to make precision measurements of unknown resistance levels.

The basic configuration appears in Fig 5.1.

The unknown resistance is R_X and R_1, R_2 and R_3 are precision resistors of known value.

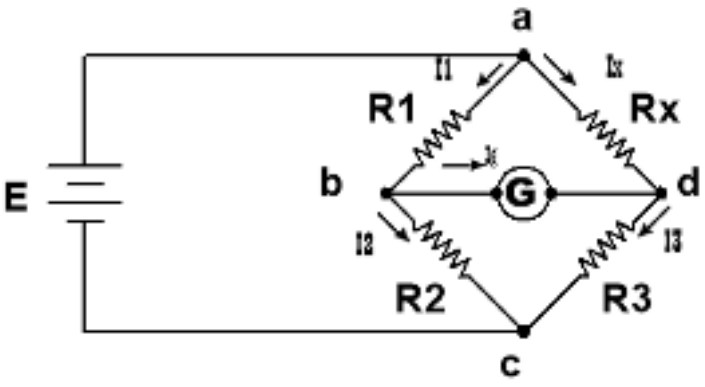


FIG 5.1

The network is balanced when the galvanometer (G) has a zero-level indication.

We are aware form circuit theory that if I_G = 0 A, the voltage V_{bd} is zero and

$$V_{ab} = V_{ad} \quad \text{and} \quad V_{bc} = V_{dc}$$

By substitution

$I_1R_1 = I_XR_X$ Eq (5.1) and $I_2R_2 = I_3R_3$ Eq (5.2)

Solving Eq (5.1) for I_1 yields.

$$I_1 = \frac{I_X R_X}{R_1}$$

Substituting I_1 for I_2 and I_x for I_3 in Eq (5.2), we have

$$I_1R_2 = I_XR_3 \quad OR \quad \frac{I_XR_X}{R_1}R_2 = I_XR_3$$

Canceling I_x from both sides and solving for R_x we obtain

$$R_X = \frac{R_1}{R_2}R_3 \tag{5.3}$$

Or in the ratio form

$$\frac{R_1}{R_2} = \frac{R_X}{R_3} \tag{5.4}$$

In the commercial Wheatstone bridge, R_1 and R_2 are variable in decade steps so that the ratio R_1/R_2 is a decimal or integral multiplier, R_3 is a continuous variable resistor such as a slider-wire rheostat.

Before the unknown resistor is connected to the terminals of the commercial bridge, the R_1/R_2 ratio (called the factor of the ratio arms) is adjusted for that particular unknown resistor.

After the resistor is connected, R_3 is adjusted until there is no delectable current indicated by the galvanometer (Galvanometer sensitivities are usually 10^{-10} A or better).

The unknown resistance value is the ratio factor times the R_3 settings.

There are certain configurations is which the resistors do not appear to be in series or parallel, under these conditions, it is necessary to convert the circuit in question from one to another.

The two circuits to be investigated in this experiment are the delta (Δ) and the wye (Y), both of which appear in Fig 5.2 to convert a Δ to Y (or vice versa), we use the following conversion equations:

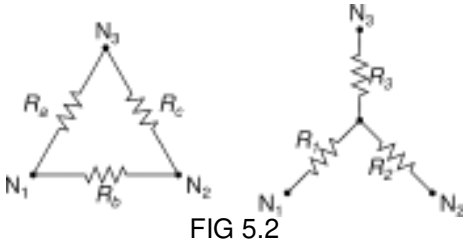


FIG 5.2

$$R_1 = \frac{R_aR_b}{R_a + R_b + R_c} \quad , \quad R_2 = \frac{R_bR_c}{R_a + R_b + R_c} \quad , \quad R_3 = \frac{R_aR_c}{R_a + R_b + R_c} \tag{5.5}$$

$$R_a = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_2} \; , R_b = \frac{R_1R_2 + R_1R_3 + R_2R_3}{R_3} \; , R_c = \frac{R_1R_2 + R_1R_2 + R_2R_3}{R_1} \tag{5.6}$$

$$IF\ R_a = R_b = R_c \qquad R_Y = \frac{R_\Delta}{3} \tag{5.7}$$

$$IF\ R_1 = R_2 = R_3 \qquad R_\Delta = R_Y * 3 \tag{5.8}$$

#PROCEDURE

- Part 1: Wheatstone bridge circuit

(a) Construct the network of Fig 5.3, insert the measured values of each resistor and set the potentiometer to the maximum resistance setting.

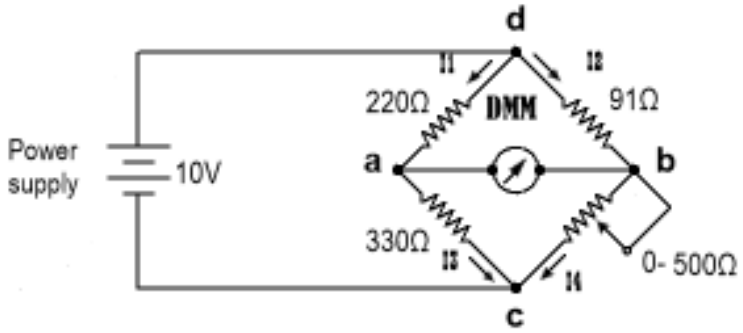


FIG 5.3

(b) Starting with the meter on a higher voltage scale, vary the potentiometer until the voltage V_{ab} is as close to zero as possible, then drop the voltage scales to the lowest range possible to set the voltage V_{ab} as close to zero volts as possible, the bridge is now balanced.

(c) Measure the voltage V_{da} , V_{db} , V_{ac} , V_{bc}.

V_{da} = ————————, V_{db} = ————————, V_{ac} = ————————,
V_{bc} = ————————.

(d) Calculate the currents I_1, I_3 and I_2, I_4 using ohm's law, are they equal as defined in the Resume of Theory?

I_1 = ————————, I_3 = ————————, I_2 = ————————,
I_4 = ————————.

(e) Disconnect one lead of the potentiometer (used as a rheostat) and measure its resistance.

R_{pot} = ————————.

(f) Verify that the following ratio is satisfied:

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

(g) Replace the 91Ω resistor by the unknown resistor, proceed as before to adjust the potentiometer until $V_{ab} \approx 0V$, remove the variable resistor and measure its resistance with the ohmmeter section of your multimeter.

R_{pot} = ————————.

(h) Calculate the unknown resistance using Eq (5.3).

R_x = ——————————.

(i) What is the maximum value of resistance that this network could measure? why?

R_{max} = ——————————.

- Part 2: Commercial Wheatstone Bridge

(a) Use the commercial Wheatstone bridge to measure the resistance of the unknown resistor.

R_x = ——————————.

(b) Measure R_x using the ohmmeter section of your multimeter, compare to the results of (a) and part 1(h).

R_x = ——————————.

- Part 3 Δ-Y Conversions

(a) Construct the network of Fig 5.4, insert the measured values of the 220Ω resistors, assume for the moment that each 1KΩ resistor is exactly 1KΩ.

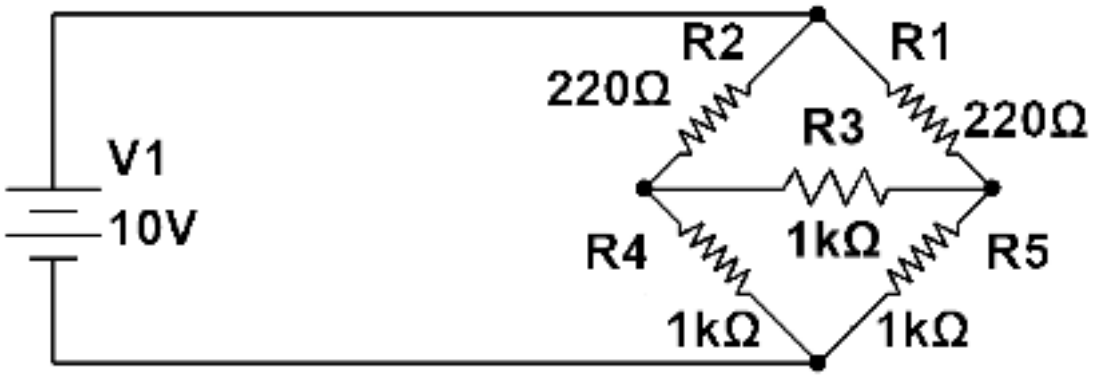


FIG 5.4

(b) Calculate the current I and the voltage V_{ab} using any method other than a Δ-Y conversion (that is, mesh analysis, nodal analysis, and so on).

I = ——————————, V_{ab} = ——————————.

(c) Measure the current I and the voltage V_{ab} and compare to the result of (b).

I = ——————————, V_{ab} = ——————————.

(d) Calculate the equivalent Y for the Δ formed by the three 1KΩ resistors, draw the equivalent circuit with the replaced by Y, insert the values of the resistors in the Y in Fig 5.5 and also indicate on diagram those available fixed resistors that have a resistance level closest to the calculated value.

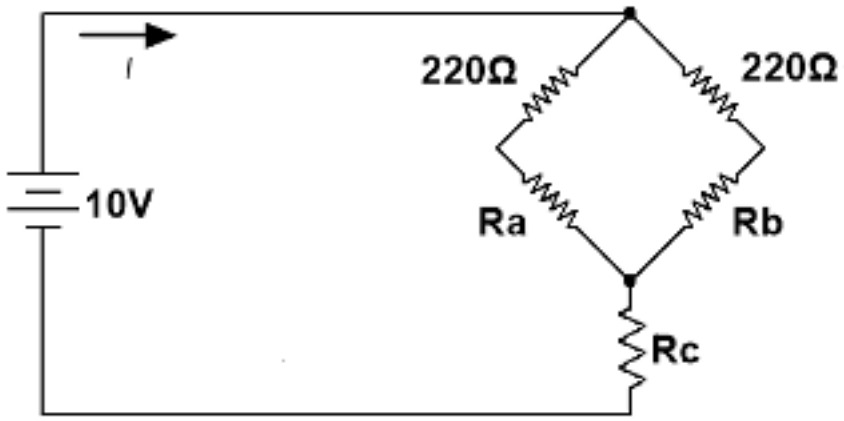


FIG 5.5

(e) Construct the network drawn on (d) and measure the current I and the voltage V_{ab} are they approximately the same as those obtained in (c)? if so, why?

I = ————————, V_{ab} = ————————.

(f) Calculate the input resistance to the network of (d) using the measured resistor value.

$R_{T\ (calculated)}$ = ————————.

(g) Disconnect the supply and measure the input resistance to the network of (d), compare the results of (f).

$R_{T\ (measured)}$ = ——————————.

(h) Determine the input resistance to the network of (a) using $R_T = E/I$ and compare to the result of (g), should they compare? Why?

R_T = ——————————————.

(i) Calculate I using the result of (f) and compare to the value measured in (e).

I = ——————————————.

The Oscilloscope

Experiment
AC

6

#OBJECT

To introduce the basic components and use of an oscilloscope and audio oscilloscope (or function generator).

EQUIPMENT REQUIRED

Instruments

1—DMM or VOM

1—Oscilloscope

1—Audio oscillator or function generator

Miscellaneous

2—D batteries with holders

#EQUIPMENT ISSUED

TABLE 5.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Oscilloscope		
Audio oscillator or function generator		

#RESUME OF THEORY

oscilloscope

The cathode-ray oscilloscope is an instrument that displays the variations of a voltage with time on a cathode-ray tube (CRT) such as that appearing in Fig 6.1, the CRT is an evacuated glass envelope, shaped as shown in Fig 6.1, with the following essential components.

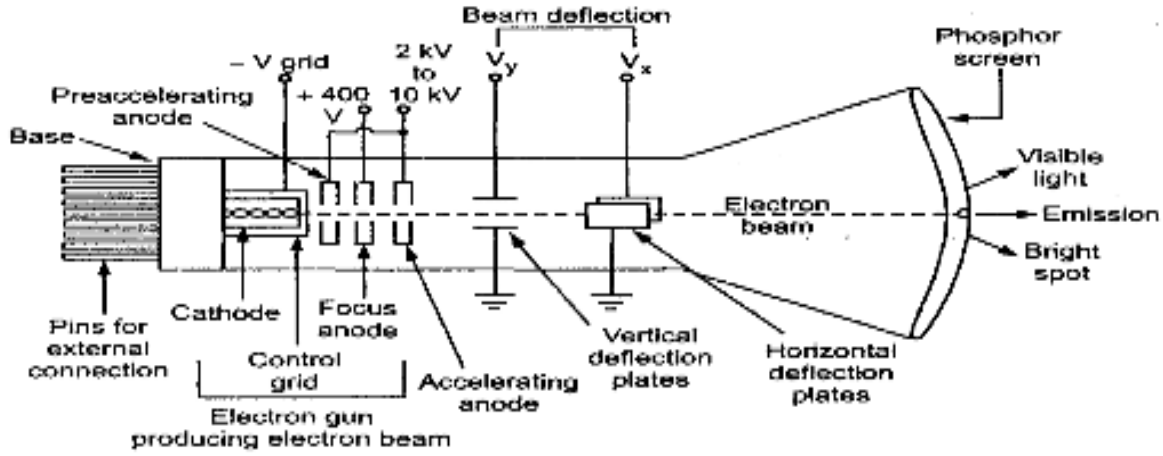


FIG 6.1

1. An electron gun generates rapidly moving electrons, shapes them into a pencil-like beam and then directs them along the axis of the tube, and it is made up from these components.
   a) A heater and cathode that serve as the source of electrons.
   b) A control electrode that serves to very the strength of the beam current.
   c) A focusing electrode that focuses the beam to sharp point on screen.
   d) Accelerating and preaccelerating electrodes that give the electrons the high velocity required to reach the screen and cause secondary emission.
2. Deflection plates for producing a vertical or horizontal deflection of the beam from its original path.
3. A fluorescent screen that emits light at the point where the beam strikes.

The deflection of the beam may be accomplished either electrostatically or electromagnetically, most oscilloscopes use electrostatic deflection (Fig 5.1), to deflect the beam, a potential is applied across plates D_1 and D_2 or D_3 and D_4, the horizontal plates D_1 and D_2 deflect the beam vertically, whereas the vertical plates D_3 and D_4 deflect the beam horizontally, the deflection of the beam is directly proportional to the impressed voltage.

The CRT is, in itself , not a very sensitive device, a 5-in CRT under ordinary condition will give about 1-in off deflection for a difference of potential of 100V, the signals encountered most frequently are usually well under 100V, so it becomes necessary to amplify the signals before a usable deflection will be obtained, two deflection amplifiers are required, one for each set of plates.

For the oscilloscope to indicate the variation of an electrical quantity in the vertical direction as function of time, there must be a voltage impressed on the horizontal deflection plates that varies linearly with the time, this voltage is shown graphically in Fig 6.2, an oscillator that can generate such a voltage is called, for obvious reasons, a sawtooth oscillator, frequently the term sweep oscillator is used, Fig 6.3 provides a block diagram of a general-purpose oscilloscope.

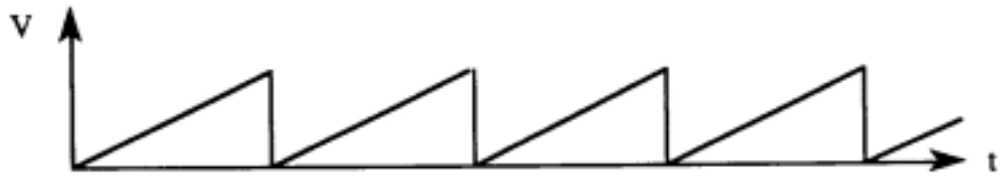


FIG 6.2

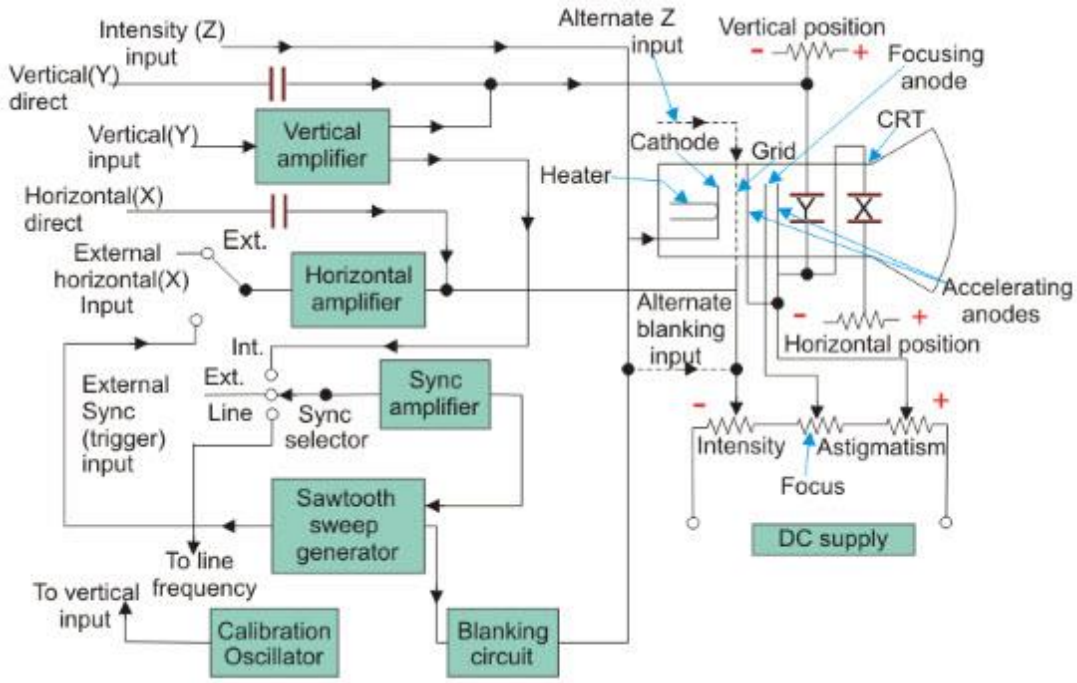


FIG 6.3

All of the controls for the proper operation of an oscilloscope are mounted on the front panel of the instrument Fig 6.4 indicates the approximate location of the controls found on most general-purpose oscilloscope.

The location of the control shown varies according to manufacturer, table 6.2 describes the function of each.

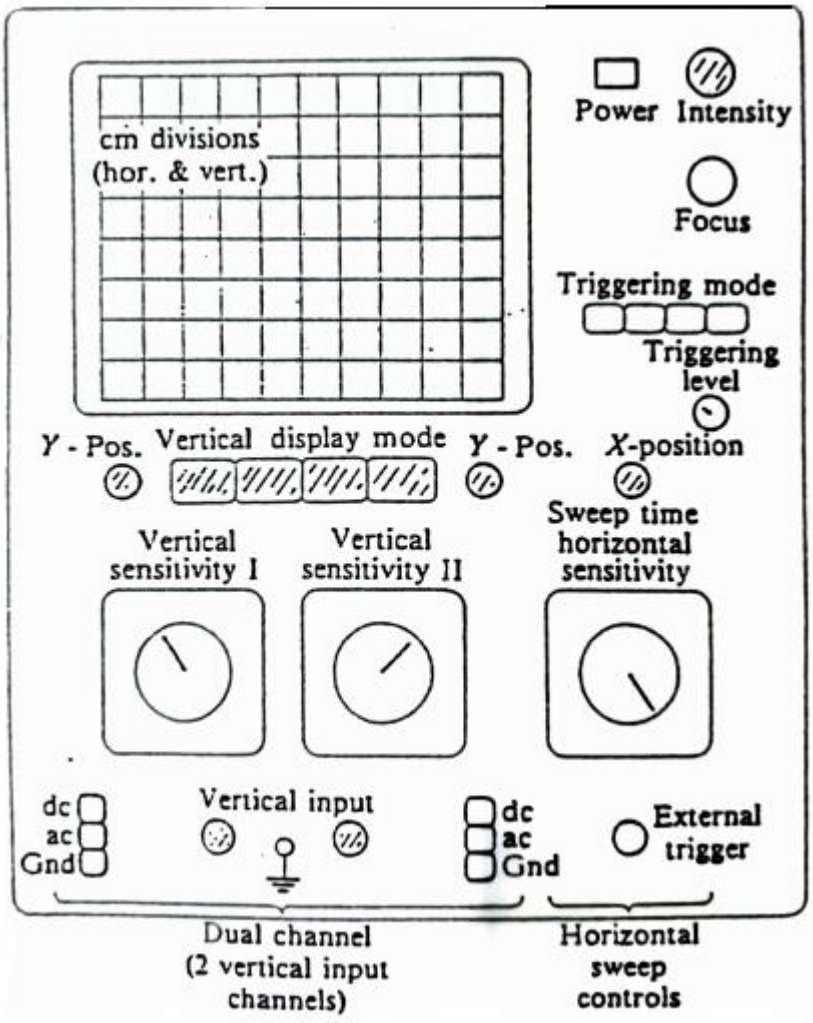


FIG 6.4

Control	Function
Power	Turns on the main power
Intensity	Controls the intensity of the pattern on the screen
Focus	Focuses the electron beam so that the pattern will be clearly defined
Triggering mode	Determines the type of triggering for the horizontal sweeping pattern
Triggering level	Determine the level at which triggering should occur
Vertical display mode	Determine whether one or two signals will be displayed at the same time and which technique will be used to display the signals
Y-position	Controls the vertical location of the pattern
X-position	Controls the horizontal location of the pattern
Vertical sensitivity	Determines the volts/cm for vertical axis of display
Sweep time horizontal sensitivity	Determines the time/cm for the horizontal axis of display
DC/AC/Gnd switch	Determines whether DC levels will be displayed on the screen and permits determining the Gnd (zero-volt input level) of the display

TABLE 6.2

(a) Voltage Measurements

DC Levels:

To use the scope to measure de levels, first place the DC/AC/Gnd switch in the Gnd position to establish the ground (zero-volt) level on the screen.

Then switch the DC/AC/Gnd switch to the DC position to measure the DC level, in the AC mode, a capacitor blocks the DC from the screen.

Then place the scope leads across the unknown DC level and use the following equation to determine the DC level:

$$DC\ level\ (V) = deflection\ (cm) * vertical\ sensitivity\ (V/cm) \qquad (6.1)$$

AC Levels:

After reestablishing the ground level, place the DC/AC/Gnd switch in the AC mode and connect the scope leads across the unknown voltage. The peak-to-peak voltage can then be determined from

$$V_{p-p}(V) = deflection\ peek\ to\ peek\ (cm) * vertical\ sensitivity\ (V/cm) \qquad (6.2)$$

(b) Frequency Measurements

The oscilloscope can be used to set the frequency of an audio oscillator or function generator using the horizontal sensitivity in the following manner.

Determine the period of the desired waveform and then calculate the number of divisions required to display the waveform on the horizontal axis using the provided μs/cm, ms/cm, or s/cm on the horizontal sensitivity control. Then adjust the audio oscillator or function generator to provide the proper horizontal deflection for the desired frequency.

Of course, the reverse of the above procedure will determine the frequency of an unknown signal.

Audio Oscillator and Function Generator

The audio oscillator is designed to provide a sinusoidal waveform in the frequency range audible by the human ear, a function generator typically expands on the capabilities of the audio oscillator by providing a square wave and triangular waveform with an increased frequency range, Either instrument is suitable for this experiment since we will be dealing only with sinusoidal waveforms in the audio range.

Most oscillators and generators require that the magnitude of the output signal be set by an oscilloscope or DMM (or VOM), That is, the amplitude dial of the oscillator is not graduated and the peak or peak-to-peak value is set by connecting the output of the oscillator to a scope or meter and adjusting the amplitude dial until the desired voltage is obtained.

PROCEDURE

Part 1: Introduction.

(a) Your instructor will introduce the basic operation of the oscilloscope and audio oscillator or function generator.
(b) Turn on the oscilloscope and establish a horizontal line centered on the Face of the screen, there are no connections to the vertical input sections of the scope for this part.
(c) Adjust the controls listed in Table 6.3 and comment on the effect.

Control	Observed Effect
Focus	
Intensity	
Y-position	
X-position	

TABLE 6.3

Part 2: DC Voltage Measurements

(a) Set the DC/AC/Gnd switch to the Gnd position and adjust the Y-position control until the zero-volt reference is a line centered vertically on the screen.
(b) Once the zero-volt level is established, move the DC/AC/Gnd switch to the de position and set the vertical sensitivity to 1V/cm and connect one channel of the scope across the 1.5-V battery as shown in Fig 6.5.

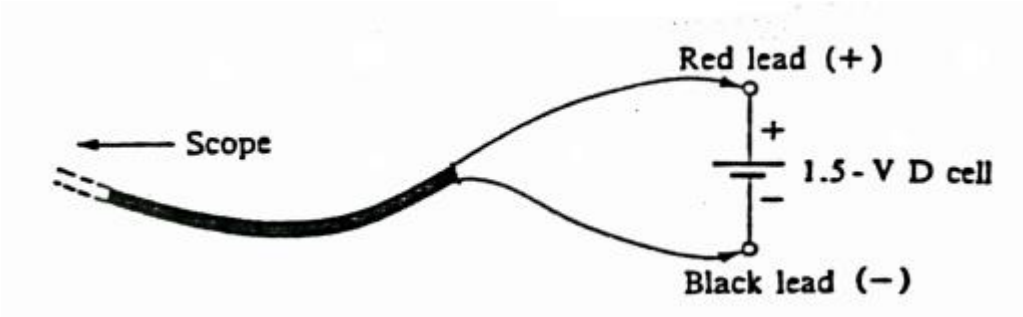


FIG 6.5

Record the vertical shift below.

Vertical shift = ———————— cm

Determine the voltage by multiplying by the vertical sensitivity, that is:

DC Voltage = (vertical shift)(vertical sensitivity)

= (————————)(————————)

= ———————— V.

Change the sensitivity to 0.5 V/cm and note the effect on the vertical shift, recalculate the DC voltage with this new shift.

DC voltage = (vertical shift)(vertical sensitivity)

= (————————)(————————————)

= ————————————V

How do the two measurements compare?

Which is more accurate? Why?

(c) Disconnect the 1.5-V battery and reestablish the zero-volts reference line, then connect the vertical input section of the scope as Fig 6.6 with the vertical sensitivity set at 1 V/cm.

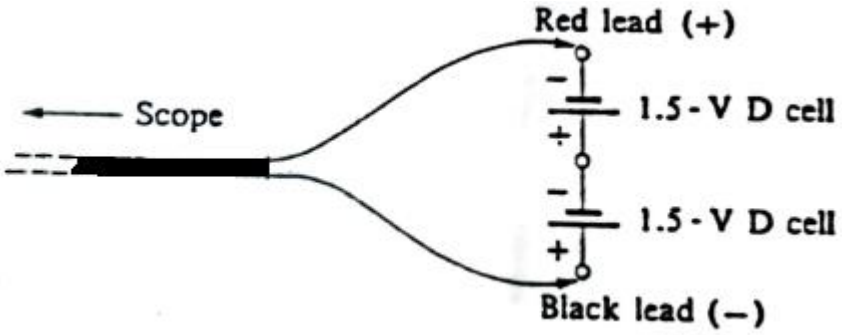


FIG 6.6

What was the effect on the magnitude and direction of the shift?

Can a scope determine whether a particular voltage is positive or negative? How?

Calculate the measured voltage as follows:

DC voltage = (vertical shift)(vertical sensitivity)

= (————————)(————————————)

= ————————————.

Measure the voltage with the DMM (or VOM) and compare results.

DC voltage [DMM or VOM] = ———————— V.

Part 3: Sinusoidal Waveforms-Magnitude.

In this part of the experiment we will learn how to set the magnitude of a sinusoid signal using an oscilloscope or DMM (or VOM), the frequency will remain fixed 500 Hz.

Oscilloscope:

(a) Connect the output of the oscillator or generator directly to one channel of the scope as shown in Fig. 6.7.

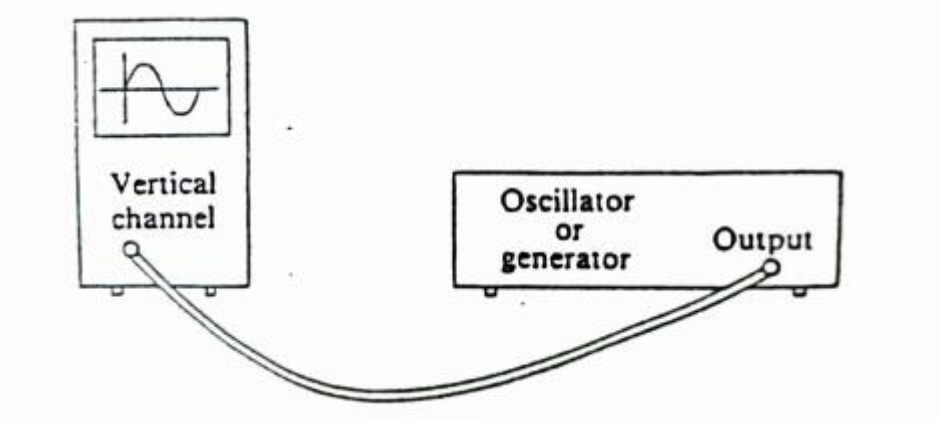


FIG 6.7

(b) Set the output frequency of the oscillator or generator to 500HZ using, the dial and appropriate multiplier, turn the amplitude knob all the way to the left for minimum output.
(c) Set the vertical sensitivity of the scope to 1 V/cm and the horizontal sensitivity to 0.5ms/cm and turn on both the scope and oscillator or generator.
(d) Set the DC/AC/Gnd switch to the Gnd position to establish the zero-volt reference level (also the vertical center of a sinusoidal waveform) and then return the: switch to the AC position.
(e) Now adjust the amplitude control of the oscillator or generator until the signal has a 4V peak-to-peak swing. The resulting waveform has the following mathematical formulation:

$$v = V_m \sin 2\Pi ft = 2 \sin 2\Pi 500t$$

(f) Switch to the de position and comment below on any change in the waveform.

(g) Make the necessary adjustments to display the following waveforms on the screen. Sketch both patterns in the space provided, showing the number of divisions (in centimeters) for the vertical and horizontal distances, and the vertical and horizontal sensitivities.

1. $v = 0.2 \sin 2\pi 500t$

2. $v = 8 \sin 2\pi 500t$

DMM or VOM:

(h) The sinusoidal signal 2 sin(2π500t) has an effective value determined by:

$$V_{eff} = 0.707V_m = 0.707(2) = 1.414V$$

Connect the DMM or VOM directly across the oscillator in the AC rms mode and adjust the oscillator output until V_{eff} = 1.414V, Then connect the output of the oscillator directly to the scope and note the total peak-to-peak swing.

Is the waveform the same as that obtained in (e)?

(i) Use the DMM (or VOM) to set the following sinusoidal output from the oscillator:

$v = 5 \sin 2\pi 500t$

V_{eff} =———————— V

Set V_{eff} with the DMM or VOM by adjusting the output of the oscillator and place the signal on the screen.

Calculate the peak-to-peak voltage as follows:

V_{p-p} =(vertical distance peak to peak)(vertical sensitivity)

=(————————)(————————)

= ———————— V

How does the above compare to the desired 10-V peak-to-peak voltage?

Part 4: Sinusoidal Waveforms-Frequency.

This section will demonstrate how the oscilloscope can be used to set the frequency output of an oscillator or generator. In other words, the scope can be used to make fir adjustments on the frequency set by the dials of the oscillator or generator.

. For a signal such as 2 sin 2=5001, the frequency is 500 Hz and the period 1/500 = 2ms. With a horizontal sensitivity of 0.5 ms/cm, the waveform should appear in exactly four horizontal divisions. If it does not, the fine-adjust control on the frequency of the oscillator or generator can be adjusted until it is exactly 4 cm. The scope has then set the output frequency of the oscillator.

Make the necessary adjustments to place the following waveforms on the scope. Sketch the waveforms in the space provided, indicating the number of vertical and horizontal deflections and the sensitivity of each.

1. $v = 0.4 \sin 60.280t$

f = ———————— Hz, t = ———————— sec.

Vertical deflection = ——————— cm.

Vertical sensitivity = ———————.

Horizontal deflection = ——————— cm.

Horizontal sensitivity = ———————.

2. $v = 5 \sin 377t$

f = ——————— Hz, t = ——————— s.

Frequency dependence of R-L-C Components

Experiment AC

7

#OBJECT

To note the effect of frequency on the basic R-L-C components.

#EQUIPMENT REQUIRED

Resistors

1—47Ω,1-kΩ

Inductors

1—10mH

Capacitors

1—0.1-μF.

Instruments

1-DMM
1--Oscilloscope
1-Audio oscillator (or function generator)

#EQUIPMENT ISSUED

TABLE 7.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Oscilloscope		
Audio oscillator or function generator		

TABLE 7.2

Resistors	
Nominal Value	Measured Value
100Ω	
1kΩ	

RESUME OF THEORY

The resistance of a carbon resistor is unaffected by frequency, except for extremely high frequencies. This rule is also true for the total resistance of resistors in series or parallel .

The reactance of an inductor is linearly dependent on the frequency applied. That is, if we double the frequency, we double the reactance, as determined by $X_L = 2\pi fL$, for very low frequencies, the reactance is correspondingly very small, while for increasing frequencies, the reactance will increase to a very large value. For dc conditions, we find that $X_L = 2\pi(0)L$ is zero ohms, corresponding with the short-circuit representation we used in our analysis of dc circuits. For very high frequencies, X_L is so high that we can often use an open-circuit approximation.

The capacitor behaves (as far as reactance is concerned) in a manner opposite to that of the inductor. For low frequencies, the capacitive reactance, as determined by $X_c = 1/2\pi fC$ is very high and approaches the open-circuit approximation used for dc circuits. For very high frequencies, the reactance becomes progressively smaller until the short-circuit approximation could be employed, the capacitive reactance is nonlinearly dependent on the frequency.

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Part 1: Resistors

Construct the circuit of Fig 7.1, insert the measured value of *R* and *Rs.*

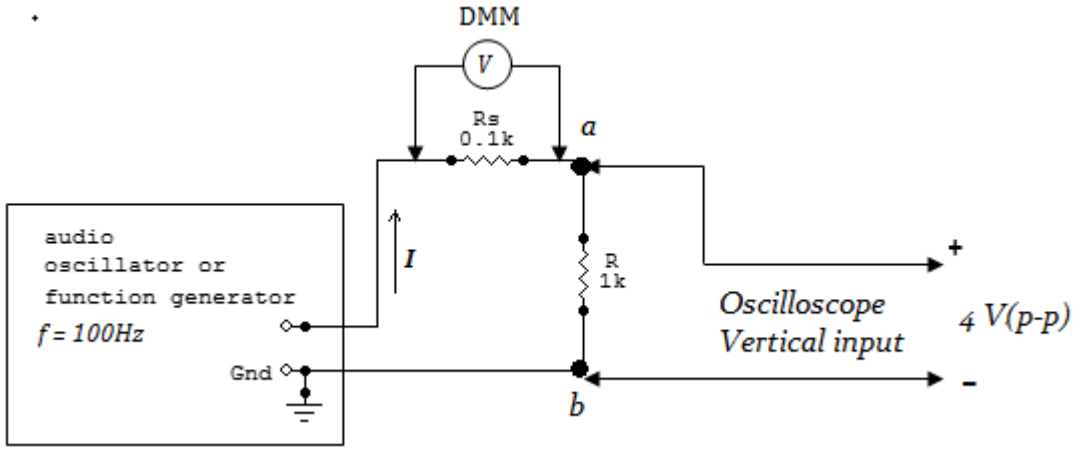


FIG 7.1

In this part of the experiment, the voltage across the resistor R will be held constant while only the frequency is varied. If the resistance is frequency independent, the current through the circuit should not change as a function of frequency. Therefore, by keeping the voltage V_{ab} constant and changing the frequency while monitoring the current I, we can verify if, indeed, resistance is frequency independent. The current I_{rms} will be determined by

$I_{rms} = VRS_{(rms)}/RS_{(measured)}$

Set the voltage Vab to 4 $V_{(p-p)}$ (with the oscilloscope). Then set the frequencies to those shown in Table 3.3, each time monitoring V_{ab} [always 4 $V_{(p-p)}$] and VRS, calculate I_{rms} and substitute into Table 7.3.

TABLE 7.3

Frequency	Vab (p-p)	$\mathrm{Vab(rms)} = \frac{0.707}{2}\mathrm{Vab(p-p)}$	I_{rms}	$\mathrm{R} = \frac{\mathrm{Vab(rms)}}{\mathrm{Irms}}$
100Hz	4V	1.1414 V		
200Hz	4V	1.1414 V		
500Hz	4V	1.1414 V		
1000Hz	4V	1.1414 V		
2000Hz	4V	1.1414 V		

Does the resistance *R* change with frequency?

Part 2: Inductors

(a) construct the circuit of Fig. 3.2 (Measure the dc resistance of the coil with the DMM and record that value as R*1*, on the diagram).

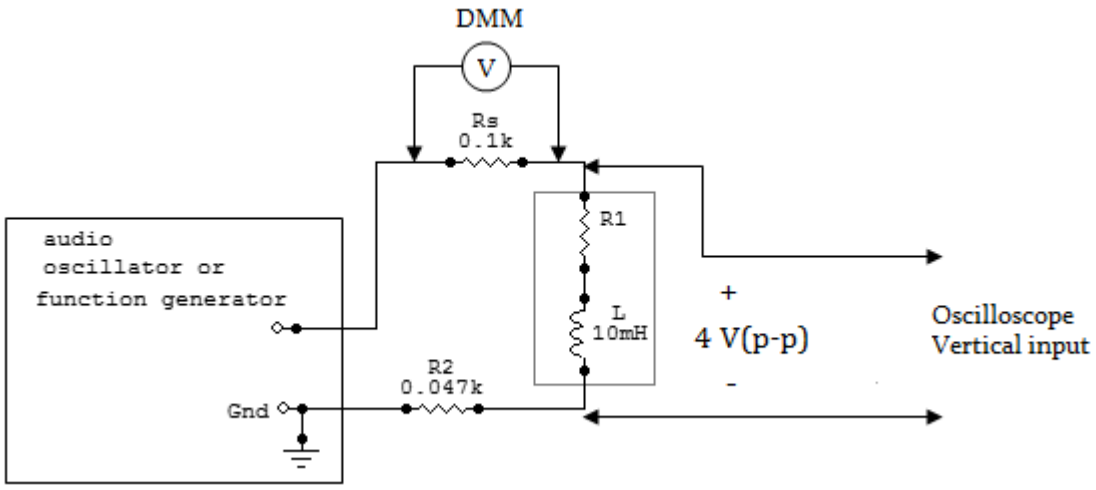


FIG.3.2

In this part, the resistor of Part 1 is replaced by the inductor, here again the voltage across the inductor will be kept constant while we vary the frequency on that voltage and monitor the current in the circuit, set the voltage V_{ab} to 4 V(p-p) (with the oscilloscope) and the oscillator or generator to the various frequencies shown in Table 3.4, each time making sure that V_{ab} = 4V(p-p) and that you measure the voltage *VRS*, Calculate I_{rms} for each frequency and substitute into Table 3.4.

TABLE 3.4

Frequency	$V_{ab(p-p)}$	$V_{ab(rms)}$	I_{rms}	$X_{L(measured)} = V_{ab(rms)}/I_{rms}$	$X_{L(calculated)} = 2\pi fL$
200Hz	4V	1.1414 V			
500Hz	4V	1.1414 V			
800Hz	4V	1.1414 V			
1000Hz	4V	1.1414 V			
2000Hz	4V	1.1414 V			

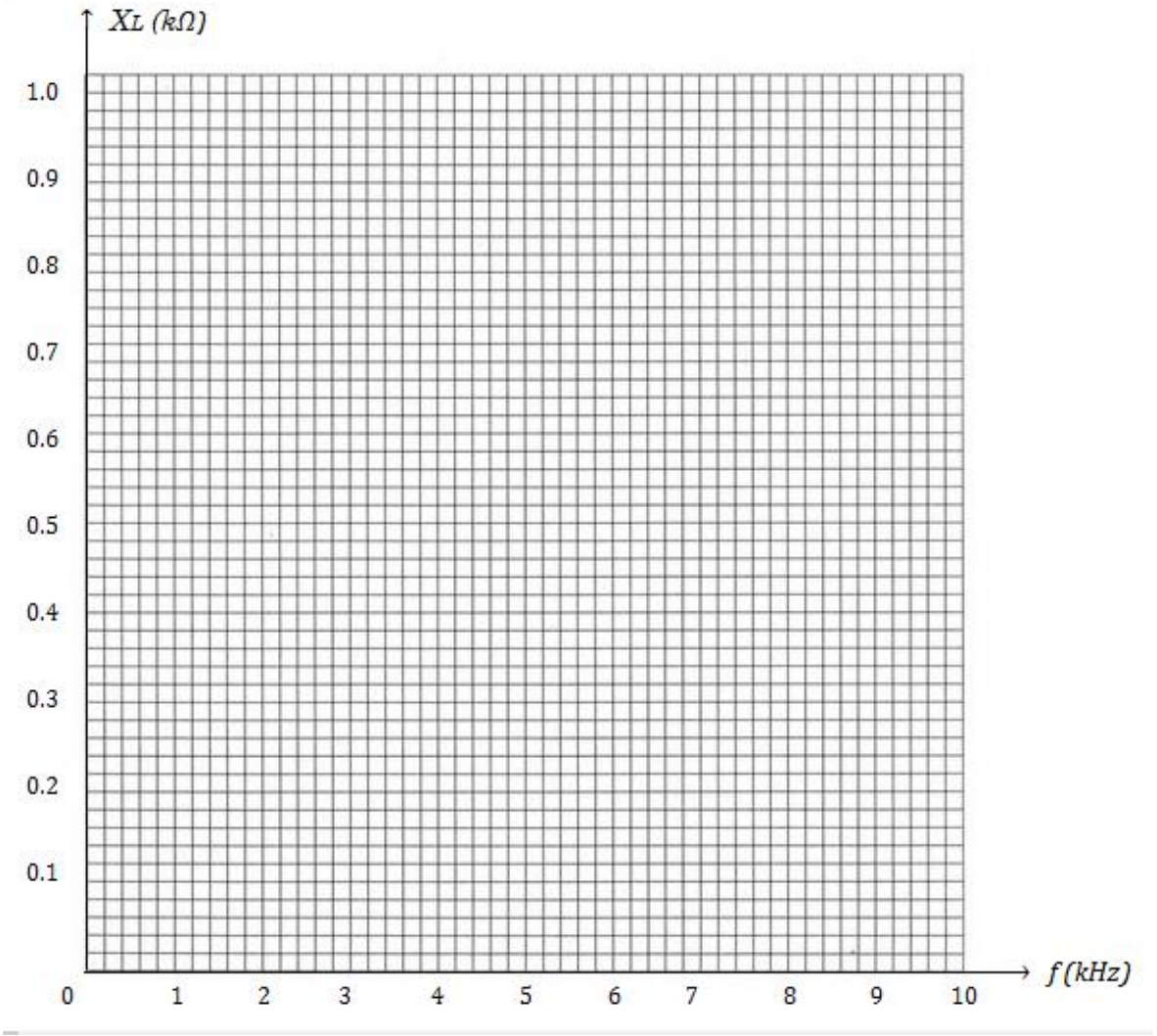
(b) Calculate the reactance XL (magnitude only) at each frequency and insert the values in Table 3.4 under the heading “XL(measured)". We are assuming that XL » R1 for the frequencies of interest, so that

$$Z = \frac{Vab(rms)}{Irms} = \sqrt{R_1^2 + X_L^2} \cong \sqrt{X_L^2} = X_L$$

(c) Calculate the reactance at each frequency of Table 3.4 using the nameplate value of inductance (10 mH), and complete the table.

(d) Plot the measured and calculated values of XL versus frequency on Graph 3.1, label each curve and plot the points accurately.

(e) Are both plots straight lines? Do they both pass through XL = 0 at f = 0Hz as they should?



GRAPH 7.1

(f) Determine the inductance at 1500 Hz using the results of (d). That is determine XL from the graph and calculate L from L = XL/2πf. Compare to the name plate value.

$L_{calculated}$ = _____, $L_{nameplate}$ = _____

Part 3 Capacitors

(a) Construct the circuit of Fig. 3.3, this part is exactly the same as Part 2 except that we now insert a capacitor in place of the inductor.

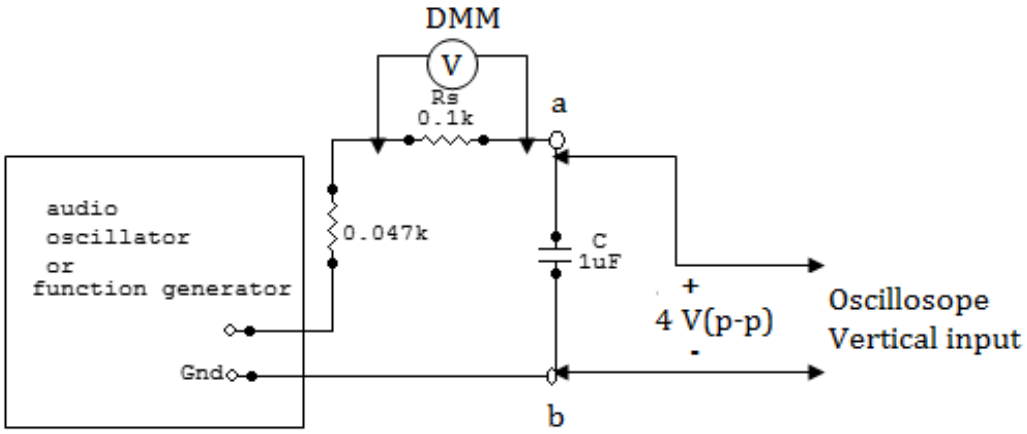


FIG 7.3

Set the voltage V_{ab} to 4 V(p-p) (with the oscilloscope) and the oscillator to the various frequencies shown in Table 7.5, make sure that V_{ab} = 4V(p-p) and that you measure voltage VRS for each setting, calculate I_{rms} at each frequency and substitute into Table 7.5.

TABLE 7.5

Frequency	$V_{ab(p\text{-}p)}$	$V_{ab(rms)}$	I_{rms}	$X_{c(mesured)} = V_{ab(rms)}/I_{rms}$	$X_{C(calculated)} = 1/2\pi fC$
200Hz	4V	1.1414 V			
500Hz	4V	1.1414 V			
800Hz	4V	1.1414 V			
1000Hz	4V	1.1414 V			
2000Hz	4V	1.1414 V			

(b) Calculate the reactance Xc (magnitude only) at each frequency and insert in Table 7.5 under the heading "$Xc_{(measured)}$".

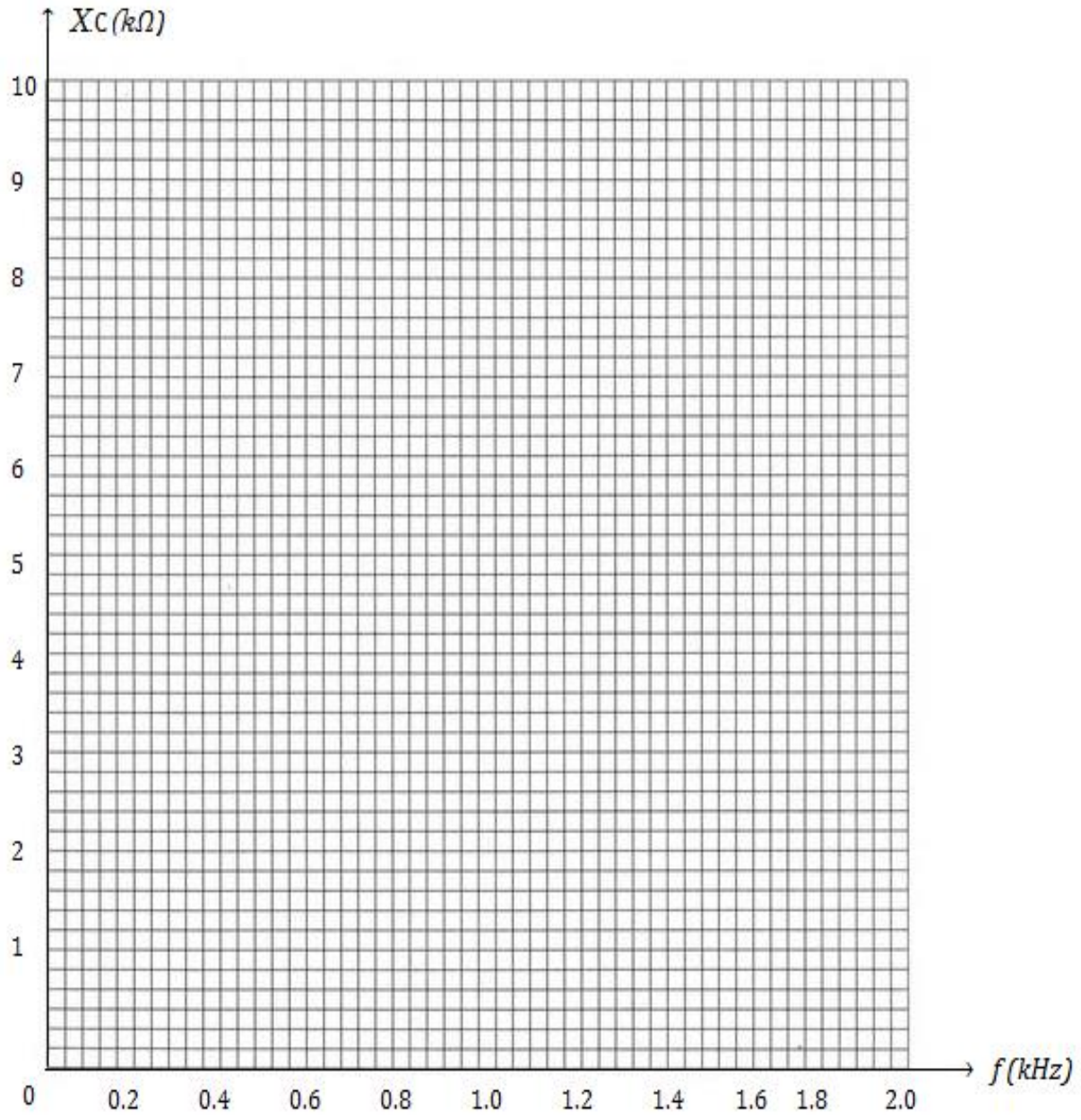
(c) Calculate the reactance at each frequency of Table 7.5 using the nameplate value of capacitance (0.1μF), and complete the table.

(d) Plot the measured and calculated values of Xc versus frequency on Graph 7.2. Label each curve and plot the points accurately.

(e) How do the two curves compare? Are they linear (straight-line) or non-linear? Comment on their shape.

(f) Determine the reactance of the capacitor at 1500 Hz from the plot and calculate the capacitance of the capacitor using the equation C = 1/2πfXc, compare to the nameplate value.

L calculated = _____, L nameplate = _____.



GRAPH 7.2

Frequency Response of R-L and R-C Networks

Experiment AC

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OBJECT:

To plot the frequency response of the R-L and R-C networks.

#EQUIPMENT REQUIRED:

Resistors:

1 100—Ω, 1-kΩ

Inductors:

1 10—mH

Capacitors:

2 0.1—μF

Instruments:

1—DMM

1—Oscilloscope

1—Audio oscillator (or function generator).

#EQUIPMENT ISSUED

TABLE 7.1

Item	Manufacturer and Model No.	Laboratory Serial No.
DMM or VOM		
Oscilloscope		
Audio oscillator or function generator		

TABLE 7.2

Resistors	
Nominal Value	Measured Value
100Ω	
1kΩ	

#RESUME OF THEORY:

For the series DC or AC circuit, the voltage drop across a particular element is directly related to its impedance as compared to the other series elements. Since the impedance of the inductor and capacitor will change with frequency, the voltage across both elements will be determined by the applied frequency.

For the series R-L network, the voltage across the coil will increase with frequency since the inductive reactance increases directly with frequency and the impedance of the resistor is essentially independent of the applied frequency.

Since the voltage and current of the resistor are related by the fixed resistance value, the shape of their curves versus frequency will be the same.

Keep in mind the voltages across the elements in an ac circuit are vectorially related.

Otherwise, the voltage readings may appear to be totally incorrect and not satisfy Kirchhoff's voltage law.

Caution: Make sure that all measurements are made with the oscilloscope and oscillator sharing a common ground. The elements of the network may have to be reversed to insure a common ground.

#PROCEDURE:

Part 1: R-L Network

(a) Construct the network of Fig. 4.1 Insert the measured values of the resistors R and R1 on the diagram. For the frequency range of interest, we will assume that $X_L >> R_1$ and $ZL = R_1 + jX_L \cong jX_L$

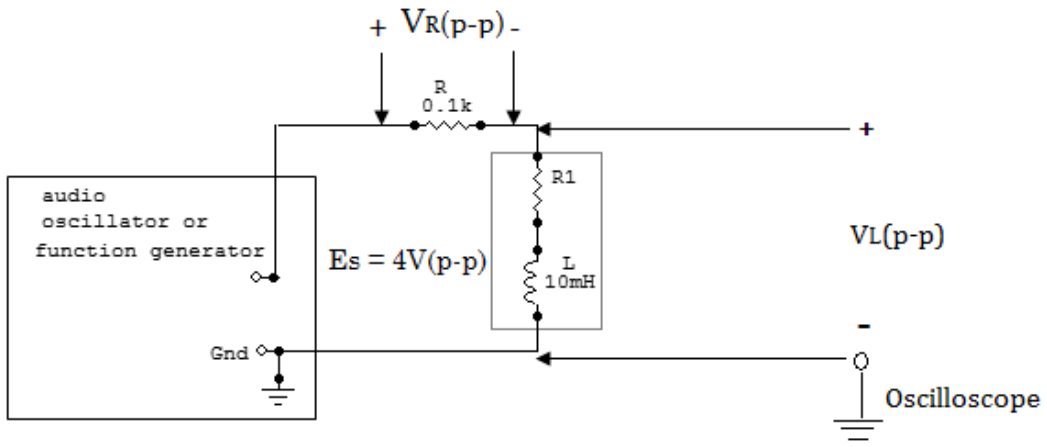


FIG 8.1

(b) Maintaining $4V_{(p-p)}$ at the input to the circuit, record the voltage $V_{L(p-p)}$ for the frequencies indicated in Table 8.3, make sure to check continually that Es = $4V_{(p-p)}$ with each frequency change.

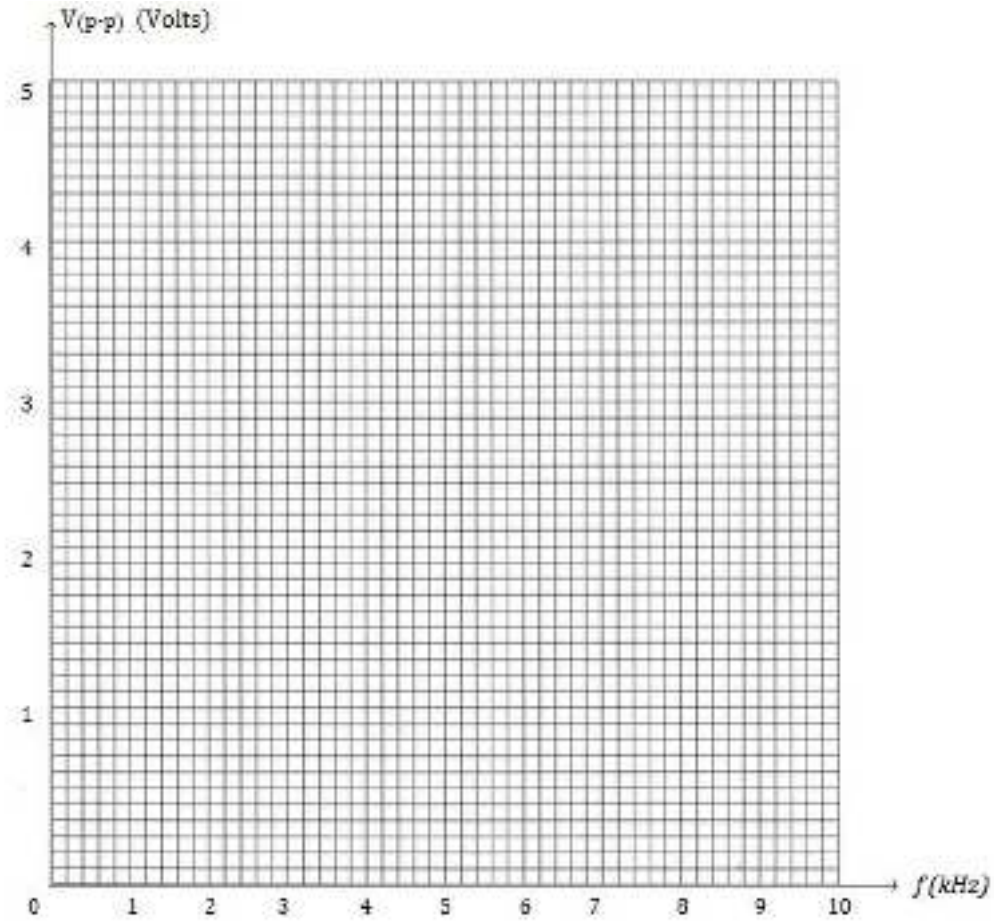
TABLE 8.3

Frequency	$V_{L(p-p)}$	$V_{R(p-p)}$	$I_{(p-p)}$
1KHz			
2KHz			
3KHz			
4KHz			
5KHz			
6KHz			
7KHz			
8KHz			
9KHz			
10kHZ			

(c) Interchange the positions of R and L in Fig 8.1 and measure $V_{R(P-P)}$ for the same range of frequencies with Es maintained at $4V_{(p-p)}$, insert the measurements in Table 8.3.

(d) Calculate $I_{(p-p)}$ from $I_{(p-p)} = V_{R(P-P)}/R$ measured and complete Table 8.3.

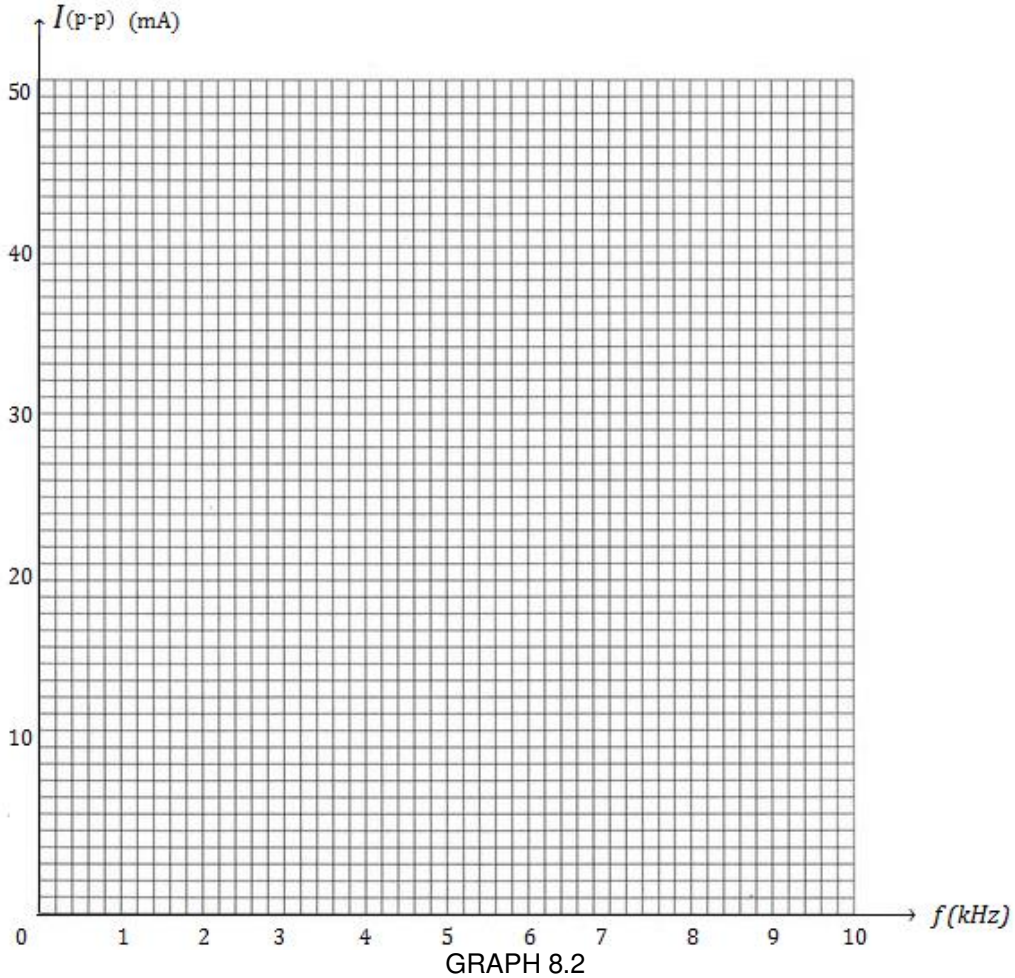
(e) Sketch the magnitude of $V_{L(p-p)}$ and $V_{R(p-p)}$ versus frequency on Graph 8.1



GRAPH 8.1

(f) At 5KHz, does the magnitude of $V_{L(p-p)} + V_{R(P-P)} = Es_{(p-p)}$? Comment accordingly, how are they related?

(g) Sketch the magnitude of $I_{(p-p)}$ versus frequency on Graph 8.2, how does it compare to the graph of $V_{R(p-p)}$ versus frequency?



GRAPH 8.2

(h) At a frequency of 8kHz, calculate the reactance of the inductor using XL= 2πfL, compare to the value obtained from the data of Table 8.3 using

$$XL \cong V_{L(p-p)}/I_{(p-p)}$$

$X_{L(calculated)}$ =_____, $X_{L(from\ data)}$ = _____.

(i) Determine, using vector algebra, the voltage VL(p-p) at a frequency of 5kHz and compare to the results appearing in Table 8.3.

$V_{L(Calculated)}$ = _____, $V_{L(measured)}$ = _____.

(j) At low frequencies the inductor approaches a low impedance short circuit equivalent and at high frequencies a high impedance open circuit equivalent, Do the data of Table 8.3 and Graphs 8.1 and 8.2 verify the above statement? Comment accordingly.

Part 2: R –C Network

(a) Construct the network of Fig 8.2 insert the measured value of the resistor R.

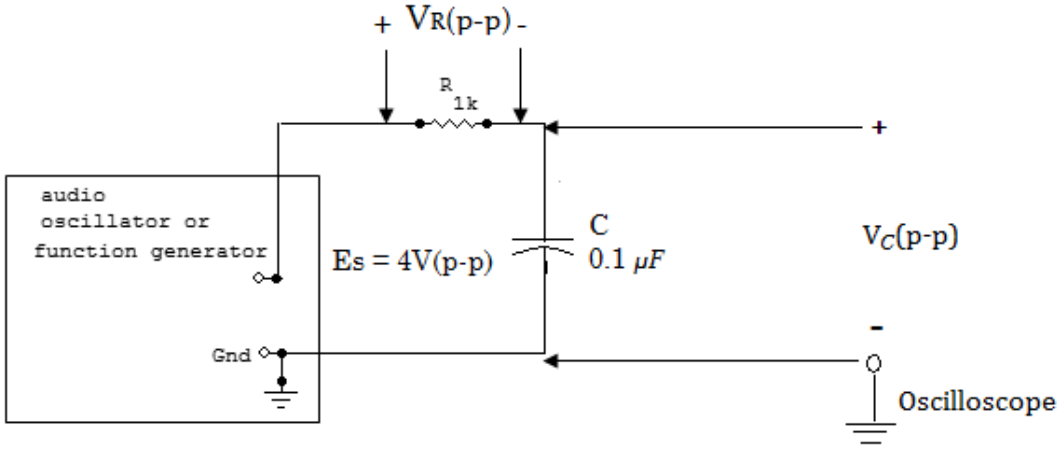


FIG 8.2

(b) Maintaining $4V_{(p-p)}$ at the input to the circuit, record the voltage $V_{C(p-p)}$ for the frequencies indicated in Table 8.4. Make sure to check continually that Es = $4V_{(p-p)}$ with each frequency change.

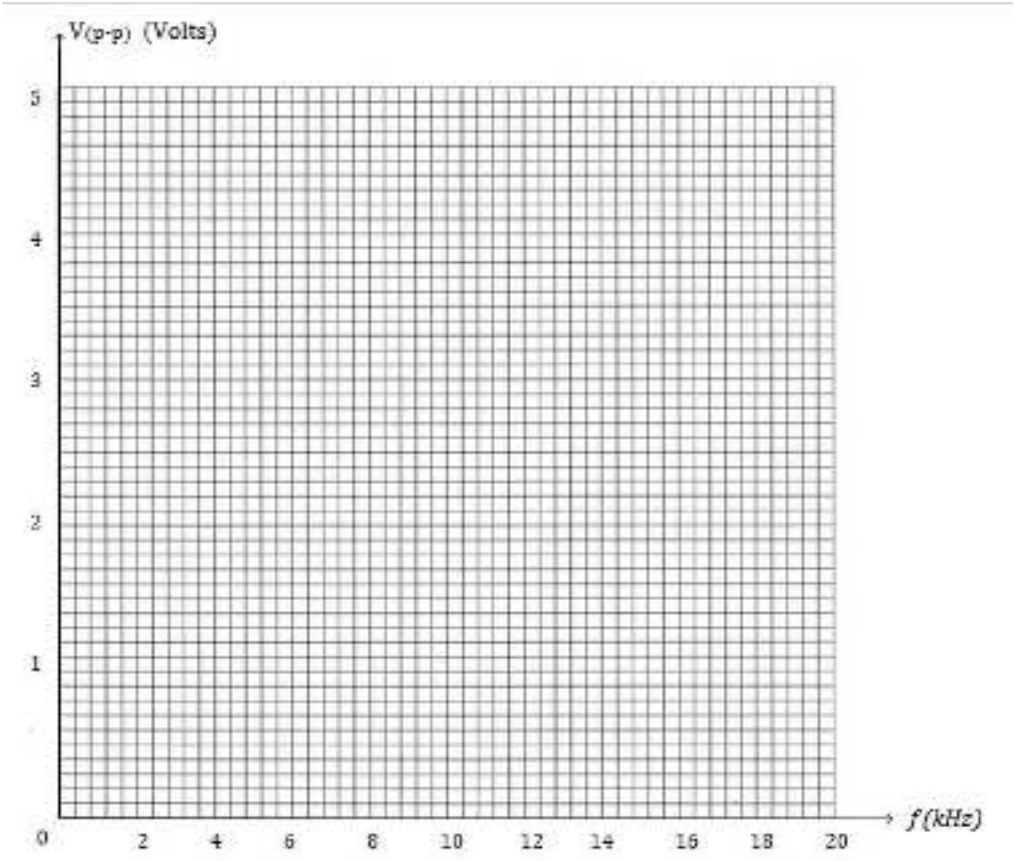
(c) Interchange the positions of R and C in Fig 8.2 and measure $V_{R(p-p)}$ for the same range of frequencies with Es maintained at $4V_{(p-p)}$, insert the measurements in table 8.4.

(d) Calculate $I_{(p-p)}$ from $I_{(p-p)} = V_{R(p-p)}/R$ measured and complete Table 8.4.

TABLE 8.4

Frequency	$V_{L(p-p)}$	$V_{R(p-p)}$	$I_{(p-p)}$
1KHz			
2KHz			
3KHz			
4KHz			
5KHz			
6KHz			
7KHz			
8KHz			
9KHz			
10kHZ			

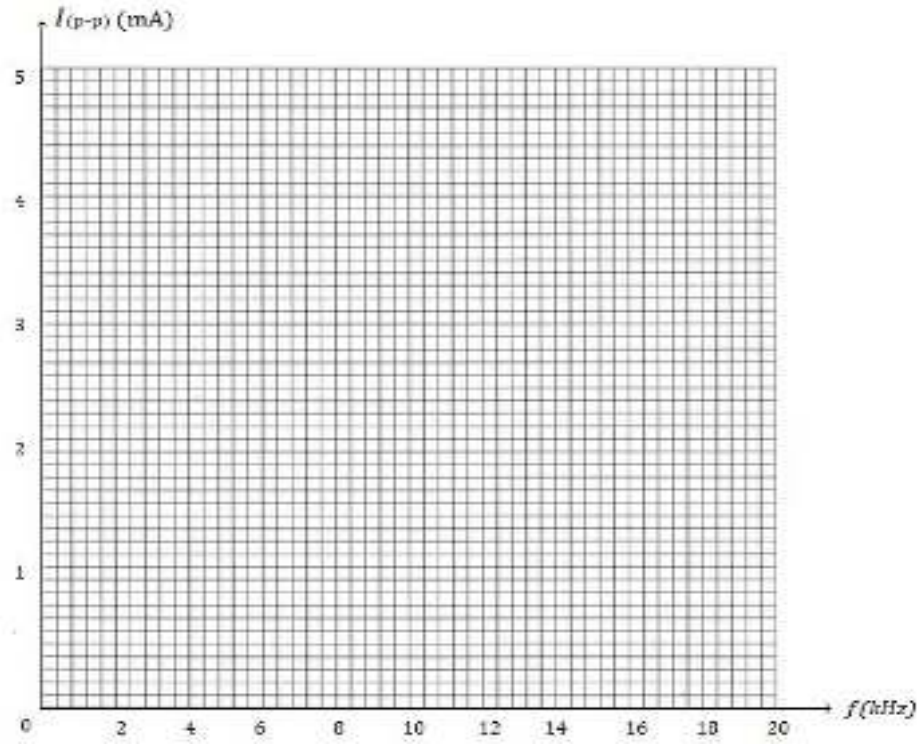
(e) Sketch the magnitude of $V_{C(p-p)}$ and $V_{R(p-p)}$ versus frequency on Graph 8.3.



GRAPH 8.3

(f) At 10 KHz, does the magnitude of $V_{L(p-p)} + V_{R(p-p)} = ES_{(p-p)}$? Comment accordingly, how are they related?

(g) Sketch the magnitude of $I_{(p-p)}$ versus frequency on Graph 8.4, how does it compare to the Graph of $V_{R(p-p)}$ versus frequency?



GRAPH 8.4

(h) At a frequency of 4kHz, calculate the reactance of the capacitor using $X_C = 1/2\pi fC$, compare to the value obtained from the data of Table 8.4 using: $X_C = V_{C(p\text{-}p)}/I_{(p\text{-}p)}$

$X_{C(calculated)}$ = _____, $X_{c\ (from\ data)}$ = _____.

(i) Determine, using vector algebra, the voltage $V_{C(p\text{-}p)}$ at a frequency of 6kHz and compare to the result appearing in Table 8.4.

$V_{C(p\text{-}p)\ (calculated)}$ = _____, $V_{C(p\text{-}p)\ (measured)}$ = _____.

(j) At low frequencies the capacitor approaches a high-impedance open-circuit equivalent and at high frequencies a low impedance short circuit equivalent. Does the data of Table 8.4 and Graphs 8.3 and 8.4 verify the above statement? Comment accordingly.

(k) Connect two 0.1 µF capacitors in parallel and plot the graph of $V_{C(p\text{-}p)}$ versus frequency on Graph 8.3. Are the results what you expected? Why? Use the same values for ES and R.

Experiment AC

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